

Branch exchange and multiplicity of the optimal excitation wavelength in nitrogen-vacancy magnetometry under pressure

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Nitrogen-vacancy centers in the culet of a diamond anvil cell are the reference local probe of megabar magnetism, and their excitation wavelength is chosen there, as at ambient pressure, to maximize the photoluminescence. That choice rests on a condition already known to be violated. Writing the lock-in sensitivity as $\eta = \Delta\nu/(C\sqrt{R})$ and assuming only that wavelength acts through the optical pumping rate $\Gamma_p \propto IA(\lambda)$, we prove three results. A Franck-Condon absorption spectrum carries a zero-phonon and a sideband branch whose weights scale differently with the Huang-Rhys factor. Compression raises that factor monotonically, so the two cross at most once and the optimum *exchanges branch discontinuously*, jumping roughly 70 nm at a pressure set by the excitation bandwidth—a dependence no other mechanism in the framework produces. The optimum is a *level set* $\{\lambda : a(\lambda) = I_c/I\}$ of exactly degenerate wavelengths, rather than a single wavelength. Its cardinality is a staircase in $\ln I$ whose steps fall at powers fixed by the *relative* absorption spectrum alone, so the ladder requires neither absolute calibration nor knowledge of which mechanism operates. The response enters η only through $E = 2c + s + 2w$, built from the power-law exponents of contrast, rate and linewidth, and the pump nonlinearity n . Splitting is exactly $En > 1$. Within the common-scale power-law response class, responses sharing (E, n) share the whole η surface, so the mechanism cannot be identified from sensitivity data at any number of powers, and contrast and linewidth must be recorded separately. The power-dependence model fitted to single nitrogen-vacancy centers at ambient pressure in the published literature corresponds to $E = 3$ and $n = 2$. We give eleven pre-specified tests, none requiring absolute calibration, and state the two spectral regions in which the reconstructed kernel used for the worked examples must not be used.

I. INTRODUCTION

The search for superconductivity in hydrogen-rich compounds, initiated by Ashcroft’s proposal that metallic hydrogen and hydrogen-dominant alloys should be high-temperature superconductors [1, 2], has produced the highest critical temperatures on record: 203 K in H₃S at 155 GPa [3] and 250–260 K in LaH₁₀ near 170 GPa [4, 5]. A second family of clathrate superhydrides operates at markedly lower pressure: CeH₉ is synthesized at 80–100 GPa and superconducts with T_c up to 115 K below one megabar [6, 7], and the substitutional alloy (La,Ce)H₉ reaches 148–178 K over 97–172 GPa [8]. Establishing that these materials are superconductors requires local magnetic evidence, and NV centers implanted a few tens of nanometers beneath the culet supply it [9–12]: the sensor is fabricated inside the anvil, sits micrometers from the sample, and survives the same pressure. The technique has been pushed from the 30 GPa scale to 130 GPa [13], to megabar pressures [14, 15], and to 240 GPa [16], and has imaged the Meissner effect and flux trapping in CeH₉ [17], in LaH₁₀ [18], in conventional superconductors under pressure [19], and in pressurized nickelates [20].

The scientific content of those experiments is carried by *maps*, and a map is a raster of optically detected magnetic resonance (ODMR) spectra whose total acquisition time scales as the number of pixels times the square of the magnetic sensitivity [21], at a fixed target field uncertainty. Sensitivity is therefore the currency in which spatial resolution, field of view, and the number of ac-

cessible (P, T) points are all purchased. One of the choices that sets it—the excitation wavelength—is normally made by maximizing the detected photoluminescence. That choice has been optimized before, but never for sensitivity: Beha *et al.* located an optimum excitation wavelength for single NV[−] centers at ambient pressure by balancing excitation against NV⁰ recharging [22], and Todenhagen and Brandt have measured how the readout contrast depends on it [23]. Both are statements about the rate or the contrast at one power. Neither is a statement about η , neither carries a power axis, and neither has been repeated under pressure. Reference [21] does not treat the excitation wavelength as an optimization variable at all, taking a single free-running 532 nm source as sufficient.

That practice rests on a condition, and the condition is testable. The continuous-wave lock-in sensitivity is $\eta \propto \Delta\nu/(C\sqrt{R})$, so the wavelength that maximizes the detected rate R minimizes η only if the ratio $C/\Delta\nu$ is stationary there. Todenhagen and Brandt have measured the excitation-wavelength dependence of the ODMR contrast at ambient pressure and find it varying between −12% and −18% across 480–575 nm [23]. That variation is not monotone—the same work reports the optical readout performing best between 525–550 nm—but as a mean logarithmic slope over that span it corresponds to about 0.43 % nm^{−1}. All the argument below requires is that the slope is nonzero. The condition for the two optima to coincide is therefore not satisfied, and the question of where the sensitivity optimum actually sits is open.

Pursuing it does not displace the optimum. It changes

what kind of object the optimum is, and establishing that structure is the purpose of this paper.

The argument rests on a single hypothesis, which we label (M) and treat as an object of measurement rather than an assumption: that the wavelength enters the response only through the optical pumping rate $\Gamma_p \propto I A(\lambda)$, so that only the number of absorbed photons matters and not which photon. Under (M) alone, with no assumption about the shape of the absorption spectrum, three results follow.

The response enters through one exponent (Theorem G, Sec. III). For power-law responses the triple $(C, R, \Delta\nu)$ reaches η only through $E = 2c + s + 2w$ and the pump nonlinearity n . Splitting is exactly the condition $En > 1$. Within that response class, models sharing (E, n) share the whole η surface, so the mechanism is unrecoverable from sensitivity data at any number of powers.

The optimum is a level set (Theorem M, Sec. IV). Given that interior maximum, the set of optimal wavelengths at power I is $\{\lambda : a(\lambda) = I_c/I\}$, its members are *exactly* degenerate in η , and its cardinality is a staircase in $\ln I$ whose steps sit at ratios of critical values of the *relative* spectrum—so the whole ladder follows from a low-power scan, with no absolute calibration and no knowledge of the mechanism.

Pressure exchanges the branch (Theorem X, Sec. V). A Franck-Condon absorption spectrum carries a zero-phonon and a sideband branch whose weights scale differently with the Huang-Rhys factor S . Compression raises S , so their ratio crosses at most once, and where it crosses the global optimum jumps roughly 70 nm between branches, at a pressure fixed by the excitation bandwidth rather than at a single number.

The two degeneracies have different causes and different signatures—the ladder is driven by power and needs $I > I_c$, the exchange is driven by pressure and exists at zero power—so they are separable in practice, and together they organize the (P, I) plane (Sec. VD). The role of pressure is thereby inverted relative to the usual framing of high-pressure sensing: rather than only a hostile environment a measurement must survive, it is the one continuously tunable variable that sweeps $dS/dP > 0$, and therefore the only accessible knob that can drive a branch exchange at all. Temperature changes the linewidths but barely changes S .

The worked examples use an absorption kernel reconstructed from the combined *ab initio* and diamond-anvil-cell study of Ref. [24]. The theorems are statements about the structure of $\eta = 1/(G\sqrt{R})$ and assume nothing about the spectrum; every number attached to them is conditional on that reconstruction. Appendix B states where it has been validated and the two spectral regions in which it must not be used.

II. FRAMEWORK AND THE TWO OPTIMA

This section fixes the two objects the rest of the paper compares—the wavelength that maximizes the photoluminescence and the wavelength that minimizes the sensitivity—states the condition under which they coincide, and states the one hypothesis, (M), on which the theorems rest. It closes with the absorption kernel used for the worked examples and with the two spectral regions in which that kernel may not be used.

A. The sensitivity functional

The scaling that follows was established for the nitrogen-vacancy magnetometer by Taylor *et al.* [25]. For continuous-wave, lock-in-detected ODMR the shot-noise-limited magnetic sensitivity is [21, 26, 27]

$$\eta_B = \frac{4}{3\sqrt{3}} \frac{h}{g_e \mu_B} \frac{\Delta\nu}{C\sqrt{R}}, \quad (1)$$

with $\Delta\nu$ the ODMR linewidth, C the resonance contrast and R the detected photon rate. Because we compare wavelengths rather than predict an absolute noise floor, we work throughout with the reduced quantity

$$\eta \equiv \frac{\Delta\nu}{C\sqrt{R}} = \frac{1}{G\sqrt{R}}, \quad G \equiv \frac{C}{\Delta\nu}, \quad (2)$$

and with its inverse square, which is the quantity all three theorems are statements about,

$$\Phi \equiv \eta^{-2} = G^2 R = \frac{C^2 R}{\Delta\nu^2}. \quad (3)$$

Every overall scale—collection efficiency, NV density, integration bandwidth—cancels from ratios of η , and none of the results below depends on the prefactor of Eq. (1).

Equation (2) does two things at once. The contrast enters linearly while the rate enters as a square root, so the two are not interchangeable and a mechanism acting on C costs four times more than one acting on R by the same factor. And η is a function of the triple $(C, R, \Delta\nu)$ only through the single combination Φ , which is the origin of Theorem G.

B. The fixed-power figure of merit

At fixed pressure and fixed *incident optical power*, the photon flux is $I/E_\gamma \propto \lambda I$, so the rate at which photons are absorbed is proportional to

$$A(\lambda) \equiv \lambda \sigma_{\text{abs}}(hc/\lambda, P), \quad a(\lambda) \equiv \frac{A(\lambda)}{A_{\text{max}}}, \quad (4)$$

and the optical pumping rate is $\Gamma_p = \gamma I A(\lambda)$ with γ a constant. The distinction between fixed incident power

and fixed incident photon flux is not cosmetic: the two conventions weight the branches of Sec. V differently by the factor $\lambda_{\text{SB}}/\lambda_{\text{ZPL}}$ and therefore return different exchange pressures. The convention is stated wherever a pressure is quoted.

We write $\ell_f \equiv d \ln f / d \lambda$ for logarithmic slopes and distinguish three wavelengths,

$$\begin{aligned} \lambda_{\text{abs}} &= \arg \max A, & \lambda_{\text{PL}} &= \arg \max R, \\ \lambda_{\eta} &= \arg \min \eta = \arg \max \Phi. \end{aligned} \quad (5)$$

Only λ_{PL} and λ_{η} are measurable, and they differ in what must be recorded: λ_{PL} needs the count rate alone, whereas λ_{η} needs rate, contrast and linewidth *separately*. Section III shows that this is a requirement of principle and not of convenience.

C. The coincidence condition

Since $\eta = 1/(G\sqrt{R})$, the sensitivity optimum maximizes $\Phi = G^2 R$ while the photoluminescence optimum maximizes R . Write ℓ_X for the logarithmic wavelength derivative of X , so that $\ell_{\Phi} = 2\ell_G + \ell_R$. Provided λ_{PL} is interior to the analysis window and R and G are differentiable there, $\ell_R(\lambda_{\text{PL}}) = 0$, and hence

$$\lambda_{\text{PL}} \text{ is stationary for } \Phi \iff \ell_G(\lambda_{\text{PL}}) = 0. \quad (6a)$$

Stationarity is not optimality, and Eq. (6a) is therefore not a statement about λ_{η} . What ties the two optima together is the stronger hypothesis that ℓ_G vanishes across the window,

$$\ell_G \equiv 0 \text{ on the window} \implies \lambda_{\eta} = \lambda_{\text{PL}}, \quad (6b)$$

and then they coincide *exactly*, at every pressure and every power, no matter how strongly the charge yield or the saturation scale depend on wavelength. The contrapositive of Eq. (6b) is what an experiment uses: an observed separation between λ_{PL} and λ_{η} is direct evidence of a wavelength-dependent $C/\Delta\nu$ and cannot be produced by the detected rate alone.

The distinction between stationarity and optimality is not academic. Case (c) of Sec. II E is a counterexample to the converse of Eq. (6a): under (M) the photoluminescence optimum sits at an interior maximum of A , so $\ell_G(\lambda_{\text{PL}}) = 0$ holds there automatically, and yet above the threshold power λ_{PL} is a local *minimum* of Φ and is not a sensitivity-optimal wavelength at all. Nor need the separation be small where it occurs. On the reconstructed kernel of Sec. II F at 120 GPa, a wavelength-independent slope ℓ_G above $0.27\% \text{ nm}^{-1}$ carries the global optimum from the sideband maximum at approximately 441 nm to the zero-phonon line at 514.5 nm, a displacement of 73.8 nm; that threshold is conditional on the reconstruction, but its order of magnitude is not.

Equation (6b) is what makes the measurement of Ref. [23] load bearing: the contrast slope quoted in Sec. I,

$\ell_G \approx 0.43\% \text{ nm}^{-1}$, gives $\ell_G \neq 0$ unless the linewidth happens to carry the same slope, and the two optima are then distinct. That proviso is not idle—no wavelength-resolved linewidth has been published—which is why ℓ_G is the object of test T3 rather than an input to the framework.

D. The mediation hypothesis (M)

All three theorems are conditional on one hypothesis.

(M) The response functions—the NV^- fraction and detection efficiency q_- , the contrast C , the linewidth $\Delta\nu$, and the saturation scale—depend on the excitation wavelength *only* through the optical pumping rate $\Gamma_p \propto I A(\lambda)$; that is, only on how many photons are absorbed and not on which photon.

Under (M) the sensitivity collapses to a function of a single variable, $\eta = h(\Gamma_p)$, and the wavelength enters only through A . This is what makes the level-set structure of Theorem M possible, and it is why we treat (M) as the object of the experimental program rather than as a background assumption.

(M) does not say that the response is trivial. It is compatible with arbitrarily strong wavelength dependence of q_- , C and $\Delta\nu$, so long as that dependence is mediated by Γ_p . It fails only for a channel that depends on the photon *energy* directly—an ionization edge, a resonance in the conduction band, an absorption band belonging to a different charge state whose envelope differs in shape from that of NV^- .

(M) is known to fail somewhere. Reference [23] reports sharp resonant features in the electrically detected magnetic resonance at 521 nm, assigned to a resonant level in the conduction band, and at 575 nm, the NV^0 zero-phonon line, where it records the largest contrast magnitude it observes. Both are photon-energy specific and neither can be produced by absorbed photon number alone. The charge-state channel that carries them also has a continuous wavelength dependence, measured in Ref. [28] and calculated from first principles in Ref. [29], so (M) is an approximation everywhere and not only at the resonances. Both features, however, lie redward of 500 nm, outside the blue window that carries the high-pressure optimum, and the same reference states that the contrast is otherwise rate limited by the same process as the photoluminescence excitation. On the present evidence the data are consistent with (M) away from the identified resonances and inconsistent with it at them—exactly the situation in which a null test of arbitrary precision is valuable. Test T6 of Sec. VI is that test.

Finally, the assumption that Γ_p is *linear* in $IA(\lambda)$ can be dropped. If Γ_p is any monotone function of $\xi = IA(\lambda)$ —as it is in Ref. [26], where the pumping rate saturates—then $\eta = h(\Gamma_p(\xi)) = \tilde{h}(\xi)$, the optimal set is

still $\{\lambda : A(\lambda) = \xi^*/I\}$, and the transition-power ratios of Theorem M are unchanged. Only the conversion from ξ to laser power becomes nonlinear.

E. Three cases under the mediation hypothesis

Under (M), $\eta = h(\Gamma_p)$ and Γ_p depends on λ only through A . Three cases exhaust them.

- (a) If R is increasing in Γ_p throughout, then $\lambda_{\text{PL}} = \lambda_{\text{abs}}$ at all powers.
- (b) If Φ is increasing in Γ_p throughout, then $\lambda_\eta = \lambda_{\text{PL}} = \lambda_{\text{abs}}$: the conventional practice is correct and there is nothing to report.
- (c) If Φ has an interior maximum at $\Gamma_p^* < \gamma I A_{\text{max}}$, then λ_η is *not a point*. It is the solution set of $A(\lambda) = \Gamma_p^*/(\gamma I)$, whose members satisfy $\eta(\lambda_i) = \eta(\lambda_j)$ exactly. Note that λ_{abs} is *not* one of them: the level $\Gamma_p^*/(\gamma I)$ lies strictly below A_{max} in this case, so the solution set brackets λ_{abs} without containing it. At λ_{abs} the pumping rate is maximal and therefore past Γ_p^* , where Φ is decreasing in Γ_p ; λ_{abs} is thus a stationary point of Φ but a local *minimum* of it, and the sensitivity there is strictly worse than at any member of the optimal set.

Case (c) is the subject of Theorem M. The degeneracy in (c) is exact—we verify it numerically to 2.2×10^{-16} —but it is not symmetric on the wavelength axis. On the reconstructed kernel the red member sits about 3 nm further from λ_{abs} than the blue member at every level, because the two flanks carry different curvature. Correspondingly the 5% penalty band at 120 GPa is 426.4–457.9 nm in the optical limit of Appendix A, which is approximately 441^{+17}_{-15} , an asymmetric interval. A test that searches for members placed symmetrically about λ_{PL} will miss them; the test must search for equal η .

F. The optical kernel is taken from outside

The theorems require no model of σ_{abs} . The worked examples do, and we take it from Ref. [24], whose combined *ab initio* and DAC study reports pressure-dependent absorption spectra, Huang-Rhys [30] factors and Debye-Waller factors to 120 GPa, by the vibronic method of Ref. [31]. At ambient pressure those quantities are to be compared with the first-principles lineshape of Ref. [32], which treats the emission side, and with the measured vibronic band at 1.945 eV of Ref. [33]; we make no quantitative comparison here. The kernel is a reconstruction of the published absorption curves; it is not an independent calculation, and we make no claim to one.

One fact forces that choice, and it is the fact on which the numerical content of Theorem M rests. A single-effective-mode Franck-Condon envelope has exactly *one*

TABLE I. The fixed-power figure of merit $A(\lambda) = \lambda \sigma_{\text{abs}}$ of the reconstructed kernel at 120 GPa, normalized to its maximum, and the optical-limit sensitivity penalty $a^{-1/2}$ each wavelength would incur. The upper block lists the interior *maxima* of A ; rungs of the multiplicity ladder sit on these, on the two interior minima and on the window edge, for which Table III gives the full set. The lower block lists commercial laser lines. These values are conditional on the reconstruction; for Theorem M what matters is the *existence* of several maxima rather than the values themselves. Positions carry a systematic ± 1 nm: an independent trace of the same source figure places the sideband maximum 0.85 nm away. Values of a carry the 1% height agreement of Appendix B 1. The zero-phonon row is the only one resting on a peak *height* rather than a position, and carries in addition the constant 11% offset against the published Debye-Waller factor measured by check K3 of Appendix B 2. The 532 nm row is an interpolation artifact quoted only to be excluded (Appendix B 3).

λ (nm)	$a = A/A_{\text{max}}$	penalty $a^{-1/2}$	assignment
<i>interior maxima of A</i>			
440.6 ^a	1	1	sideband, global max
475.6	0.660	1.231	sideband
500.2	0.277	1.899	sideband
514.5	0.694–0.780	1.13–1.20	zero-phonon line
<i>commercial laser lines</i>			
405	0.586	1.306	at the ionization edge
445	0.992	1.004	inside the 5% band
457	0.916	1.045	inside the 5% band
473	0.688	1.205	outside
488	0.423	1.538	outside
505	0.110	3.02	outside
532	0.006	13	interpolated only

^aNumerical maximum of the interpolant; reported in prose as approximately 441 nm, not as sub-nanometre source precision.

interior local maximum, at every pressure from ambient to 120 GPa, and never splits. The reconstructed kernel has *four* at 120 GPa (Table I). Every rung of the multiplicity ladder lives on the critical points and edges of such a kernel, so no single-mode envelope can produce them at any parameter value. This is a statement about structure, not about goodness of fit, and it should not be argued from a fit score. Appendix B sets out where the reconstruction has been validated and where it has not.

III. THE RESPONSE EXPONENT

Theorem M, proved in the next section, needs an antecedent: the sensitivity must have an interior optimum in the pumping rate at all. This section supplies it, and finds that the antecedent is both easier to meet and more consequential than it looks. Easier, because the whole response enters through one number, so a single measured exponent decides it. More consequential, because that same collapse makes the mechanism unrecoverable from the sensitivity.

Theorem G. η depends on the response triple $(C, R, \Delta\nu)$ only through $\Phi = C^2 R / \Delta\nu^2$. For power-law responses

$$R = \Gamma_p (1 + \rho)^{-s}, \quad C = (1 + \rho)^{-c}, \quad \Delta\nu = (1 + \rho)^w, \quad (7)$$

written in terms of the reduced variable

$$\rho = (\Gamma_p / \Gamma)^n, \quad (8)$$

with Γ a common response scale and n the exponent with which the response variable follows the pumping rate, one has

$$\Phi = \Gamma_p (1 + \rho)^{-E}, \quad \boxed{E \equiv 2c + s + 2w}, \quad (9)$$

so that Φ has an interior maximum if and only if $En > 1$, and then

$$\rho^* = \left(\frac{\Gamma_p^*}{\Gamma} \right)^n = \frac{1}{En - 1}. \quad (10)$$

The proof is a substitution: $\Phi = (1 + \rho)^{-2c} \Gamma_p (1 + \rho)^{-s} (1 + \rho)^{-2w} = \Gamma_p (1 + \rho)^{-E}$, whence $d \ln \Phi / d \ln \Gamma_p = 1 - En \rho / (1 + \rho)$, which vanishes at $\rho^* = 1 / (En - 1)$ and only for $En > 1$. ■

The response therefore enters η through the pair (E, n) and nothing else. The exponent n is a property of how the response variable is driven rather than of which response is affected: $n = 1$ is the linear case, and Sec. III A exhibits a published fit with $n = 2$.

Table II evaluates E for the standard CW-ODMR mechanisms at $n = 1$. The content of the table is that the question “which mechanisms can split the optima?” collapses to the sign of $En - 1$. Rate saturation alone and linewidth power broadening alone both give $E = 1$ and cannot split at $n = 1$. Saturation together with broadening gives $E = 2$ and can. Contrast collapse alone also gives $E = 2$ and can—because η is first order in C and only half order in R , which is the structural asymmetry noted in Sec. II A. All three together give $E = 4$ and $\rho^* = 1/3$.

Two further restrictions bound the degeneracy. It holds at fixed n : two mechanisms with the same E but different n have different ρ^* and different Φ , and at $n = 2$ the rate-saturation row of Table II does split. And it holds within one functional class, since Φ sees the whole response and not only its exponents; a response outside the form of Eq. (7) with the same nominal (E, n) is not degenerate with these.

The composite rows of Table II require the mechanisms to share a common half-scale Γ ; with distinct scales E is not a single scalar, and the gauge degeneracy of Sec. III B is exact only in the regime where one scale dominates or the scales coincide.

A. The antecedent $En > 1$ is a measured quantity

Theorem M requires an interior maximum of Φ , and Theorem G reduces that requirement to $En > 1$. That

TABLE II. The splitting exponent $E = 2c + s + 2w$ for the standard CW-ODMR response mechanisms, evaluated at $n = 1$ and with a common response scale. Splitting—and therefore the entire level-set structure of Theorem M—is exactly the condition $En > 1$. The last two columns are what η can see; the mechanism labels in the first column are what it cannot (Sec. III B). The final row is the published fit of Ref. [26], which carries $n = 2$ (Sec. III A).

mechanism	c	s	w	E	n	splits	ρ^*
rate saturation alone	0	1	0	1	1	no	—
power broadening alone	0	0	1/2	1	1	no	—
saturation + broadening	0	1	1/2	2	1	yes	1
contrast collapse alone	1	0	0	2	1	yes	1
all three	1	1	1/2	4	1	yes	1/3
Ref. [26]	1	0	1/2	3	2	yes	1/5

requirement is not assumed here. Reference [26] measured R , C and $\Delta\nu$ against power for single NV centers at ambient pressure and fitted them to closed forms; E and n are read off those forms, and give $En = 6$.

Reference [26] reports, for single NV centers, the closed forms

$$R = R_\infty x, \quad C = \Theta \frac{1}{1 + \rho}, \quad \Delta\nu = \frac{\Gamma_c^\infty}{2\pi} \sqrt{\kappa} \sqrt{1 + \rho}, \quad (11)$$

in which $\varsigma = P_{\text{opt}} / P_{\text{sat}}$ is the optical pumping power normalized to saturation, written ς to keep it distinct from the rate exponent s , and $x = \varsigma / (1 + \varsigma)$ is the resulting saturated pumping rate. The remaining symbols are the saturated contrast Θ and rate R_∞ , the saturated pumping and coherence rates Γ_p^∞ and Γ_c^∞ , the Rabi frequency Ω_R , and the combinations $\kappa = \Omega_R^2 / (\Gamma_p^\infty \Gamma_c^\infty)$ and $\rho = x^2 / \kappa$.

The identification with Eq. (7) is through x , not through ς : it is x that plays the role of Γ_p , because R is exactly linear in it. This is the nonlinear-pumping case admitted in Sec. II D, and it is why the exponent n of Eq. (8) is needed. Reading off the exponents gives $c = 1$, $s = 0$ and $w = \frac{1}{2}$, so $E = 3$; and because ρ is quadratic in x rather than linear, $n = 2$ and $En = 6$, whence

$$\rho^* = \frac{1}{En - 1} = \frac{1}{5}. \quad (12)$$

The numerical maximizer reproduces that identity to five decimals, at $x^* = \sqrt{\kappa/5} = 0.067$ and $\varsigma^* = 0.072$, i.e. at 7% of the saturation power for the representative drive $\Omega_R = 3 \times 10^6 \text{ rad s}^{-1}$.

The *splitting condition* is drive independent: $\rho^* = 1/5$ is identical for all five microwave drive strengths Ref. [26] reports ($\Omega_R = 1.1, 2.5, 3.0, 3.6$ and $5.5 \times 10^6 \text{ rad s}^{-1}$), so it is a property of the functional form and cannot be removed by microwave tuning. The *power at which the optimum falls* is not: since $x^* = \sqrt{\kappa/5}$ and $\kappa \propto \Omega_R^2$, ς^* scales with Ω_R , running from 0.025 to 0.14 across those same five settings. Any statement about where in the power sweep the optimum sits must carry the drive strength with it.

Reference [26] plots that optimum and describes it in words, so the interior maximum of Φ was observed in 2011. What Theorem M adds is that under (M) this single fact makes the optimal set a level set, degenerate and multi-valued, with a ladder fixed by the relative spectrum.

B. The mechanism is not identifiable from the sensitivity

At fixed n , the models with a given E form a plane $2c + s + 2w = E$ in exponent space, and every model on that plane has an identical Φ . We verify the degeneracy numerically across the plane $E = 2$ at $n = 1$: the maximum relative difference in Φ between members is 6.5×10^{-16} [Fig. 1(a)]. It follows that the entire $\eta(\lambda, I)$ surface, the value of Γ_p^* , the critical intensity $I_c \equiv \Gamma_p^*/(\gamma A_{\max})$ defined in Eq. (13) below, and the whole multiplicity ladder are identical for all of them.

Mechanism attribution from wavelength scans is therefore impossible, at any number of powers, within the common-scale power-law response class. An observed split does not identify contrast collapse as its cause, because saturation combined with power broadening produces exactly the same η surface. What separates them is the individual responses at a given pumping rate: at $\Gamma_p = 1$, contrast collapse gives $(R, C, \Delta\nu) = (1.000, 0.500, 1.000)$ while saturation with broadening gives $(0.500, 1.000, 1.414)$ [Fig. 1(b)]. This yields a hierarchy of identifiability:

measured	determined
η alone	$\Phi = C^2 R / \Delta\nu^2$ only (degenerate)
$+ R$	$G = C / \Delta\nu$
$+ C$ or $\Delta\nu$	all of $(C, R, \Delta\nu)$

Recording C and $\Delta\nu$ separately is therefore gauge fixing, not laboratory hygiene: an experiment that records only the sensitivity, or only the sensitivity and the rate, *cannot in principle* identify the mechanism at any precision. None of this reaches the ladder, however: its transition powers depend on the response only through Γ_p^* , so the degeneracy that defeats mechanism attribution leaves the prediction of the next section untouched.

IV. LEVEL SET AND MULTIPLICITY LADDER

Case (c) of Sec. II E is the interesting one, and Sec. III A has just shown that published data put us in it. What has never been asked is what the solution set of that case looks like. It is not a point, and its size is an observable.

Theorem M. Assume (M), and let Φ be continuous with $\Phi(\Gamma_p) \rightarrow 0$ as $\Gamma_p \rightarrow 0^+$ and with a *unique interior local maximum*, at Γ_p^* —which by Theorem G exists exactly

when $En > 1$, and which Sec. III A shows is measured. Assume further that A is continuous and piecewise C^1 on the closed analysis window $[\lambda_b, \lambda_r]$, with finitely many strict local extrema, with its interior critical values pairwise distinct and distinct from $A(\lambda_b)$ and $A(\lambda_r)$, and strictly monotone in a neighborhood of each endpoint. Then the set of sensitivity-optimal wavelengths at incident power I is the level set

$$\Lambda_\eta(I) = \left\{ \lambda : a(\lambda) = \frac{I_c}{I} \right\}, \quad I_c \equiv \frac{\Gamma_p^*}{\gamma A_{\max}}. \quad (13)$$

Its cardinality $N(I) = |\Lambda_\eta(I)|$ is piecewise constant in $\ln I$ and changes only where the level I_c/I crosses a critical value of a , by

$$\Delta N = \begin{cases} +2, & \text{interior maximum,} \\ -2, & \text{interior minimum,} \\ -1, & \text{edge, } A \text{ increasing inward,} \\ +1, & \text{edge, } A \text{ decreasing inward,} \end{cases} \quad (14)$$

and the transition powers are

$$\boxed{\frac{I_k}{I_c} = \frac{1}{a_k} = \frac{A_{\max}}{A_k}} \quad (15)$$

with A_k the critical values of A .

Proof. Under (M), η is a function $h(\Gamma_p)$ of the pumping rate alone, and Γ_p depends on λ only through $A(\lambda)$. Hence

$$\arg \min_\lambda \eta = \{ \lambda : \Gamma_p(\lambda) = \Gamma_p^* \} = \{ \lambda : A(\lambda) = \Gamma_p^*/(\gamma I) \}, \quad (16)$$

which is Eq. (13). A continuous Φ that vanishes at the origin and has a single interior local maximum is strictly increasing on $(0, \Gamma_p^*)$; hence for $I < I_c$ the required level exceeds $A_{\max}$, is unattainable, and the constrained minimum is the single point λ_{abs} . For $I > I_c$ the level $a^* = I_c/I$ decreases monotonically as I rises.

The counting is elementary and does not need A to be smooth, which matters because the kernel used later is a piecewise-linear interpolant and so has no nondegenerate critical points in the sense of Morse theory. Between consecutive critical values, A is strictly monotone on each of the finitely many subintervals into which its extrema divide the window, so $A^{-1}(y)$ contains exactly one point per subinterval that spans y , and that count cannot change. As y descends through a strict interior maximum, the two subintervals meeting there begin to span y together and two preimages appear; through a strict interior minimum, two merge and disappear. Pairwise distinct critical values are what keeps these events separate, so that the observable step is ± 2 rather than a superposition. At an endpoint λ_e , where A is monotone, a single branch exists for y on one side of $A(\lambda_e)$

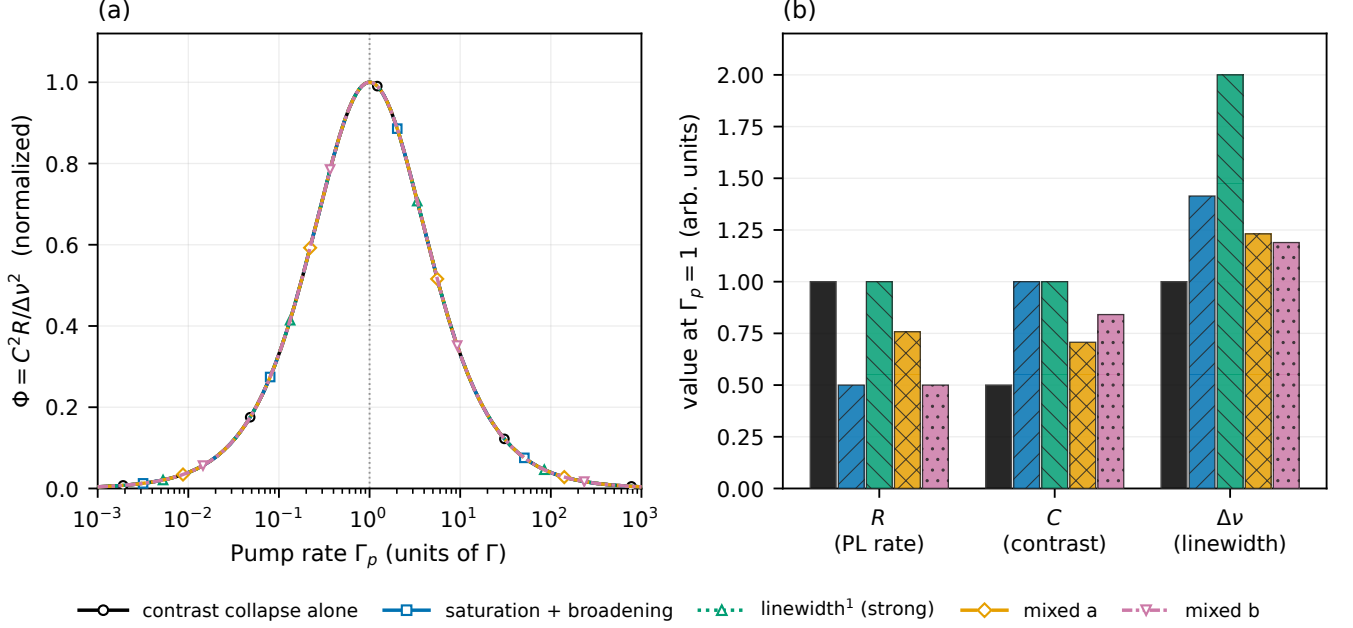


FIG. 1. The gauge degeneracy of Theorem G. (a) The reduced figure of merit $\Phi = C^2 R / \Delta \nu^2$, normalized to its maximum, against the optical pumping rate Γ_p in units of the half-scale Γ , on a logarithmic abscissa, for five representatives of the plane $E = 2c + s + 2w = 2$ at $n = 1$, two of which appear in Table II. The five curves are drawn and are indistinguishable: the largest relative difference anywhere on the plane is 6.5×10^{-16} , so no sensitivity measurement at any number of powers can separate them. (b) The three individually measurable factors for the same five models at $\Gamma_p = \Gamma$, in arbitrary units in which the first model has $R = \Delta \nu = 1$: detected rate R , contrast C and linewidth $\Delta \nu$, which span a factor 2 across the five. Separating the mechanisms requires recording C and $\Delta \nu$ apart; measuring η better cannot do it.

and leaves the domain on the other: one preimage is lost if A increases into the interior and one is gained if it decreases, which is the last two lines of Eq. (14). Both edges of the window used in Sec. IV B increase inward, so both contribute -1 there. Setting $a^* = a_k$ gives $I_k = I_c / a_k$, which is Eq. (15). ■

For a smooth A the same count is the regular interval theorem together with the Morse lemma [34, Thm. 3.1]; the piecewise argument above is given because the reconstruction of Sec. III F is not smooth, and because it is the argument the released code implements.

A. The ladder is calibration free

The practical content of Eq. (15) is in the ratios. For any two rungs,

$$\frac{I_k}{I_j} = \frac{a_j}{a_k}, \quad (17)$$

in which Γ_p^* , γ and every response parameter cancel identically.

Reading the a_k off a measurement needs one condition beyond (M), which constrains the wavelength dependence and says nothing about the behavior of the response near zero power:

(L) In the low-excitation limit the detected rate is linear in the pumping rate with no

offset, $R(\Gamma_p) = R'(0) \Gamma_p + o(\Gamma_p)$ with $R'(0) > 0$, after subtraction of background and dark counts.

Hypothesis (L) is what “unsaturated” means operationally, and it is independent of (M). Under (M) and (L) together, $R \propto A$ at low power, so the critical values a_k are read directly off a low-power photoluminescence scan.

That scan must be taken *at fixed incident optical power*, the convention of Sec. II B. A scan holding the incident photon flux fixed instead returns σ_{abs} rather than $A = \lambda \sigma_{\text{abs}}$; its critical values are those of a different function, and the rung ratios of Eq. (17) are then wrong by λ_k / λ_j , which reaches 29% across the analysis window—an order of magnitude beyond the uncertainty the a_k otherwise carry.

The positions of the rungs are therefore predicted from a *measured relative spectrum alone*, with no absolute power calibration and no knowledge of which mechanism operates. Only the overall scale I_c has to be supplied, and a single-wavelength power sweep supplies it (test T4 of Sec. VI).

This is the sense in which Theorem M functions as a prediction rather than a fit. Nothing is adjusted between the low-power scan and the observed step positions.

TABLE III. The multiplicity ladder of the reconstructed kernel at 120 GPa, enumerated over $1 < I/I_c \leq 6$. The index k numbers the rungs in order of power, and is the numeral each vertical line carries in Fig. 2(b); rungs 2 and 3 lie 0.3% apart in power and share one numeral there. “Observed” counts the members of $\Lambda_\eta(I)$ found by direct enumeration; “predicted” is Eq. (15) evaluated from the critical values of a alone. The six agree to better than 0.01%, which checks the enumeration rather than the values; those carry about $\pm 2\%$ from the 1% height reconstruction of Appendix B 1, and their fourth and fifth digits are set by the 0.005 nm critical-point grid. The listing is truncated at $I/I_c = 6$ and further rungs exist above it. The first row inherits the zero-phonon uncertainty of Table I and may lie anywhere in $1.28 \leq I/I_c \leq 1.44$; it remains the lowest rung across that range, so the N and ΔN columns are unchanged by it. The values are conditional on the reconstruction; what Theorem M freezes is the ladder *structure*.

k	I_k/I_c obs.	I_k/I_c pred.	from (nm)	kind	N	ΔN
1	1.44	1.44	514.5	maximum	$2 \rightarrow 4$	+2
2	1.51	1.51	475.6	maximum	$4 \rightarrow 6$	+2
3	1.52	1.52	474.5	minimum	$6 \rightarrow 4$	-2
4	1.86	1.86	402.0	edge ^a	$4 \rightarrow 3$	-1
5	3.61	3.61	500.2	maximum	$3 \rightarrow 5$	+2
6	4.30	4.30	497.9	minimum	$5 \rightarrow 3$	-2

^a Not a critical point of A . This rung is generated by the blue edge of the analysis window, which is where the source figure stops rather than where the absorption band ends (Appendix B 2, K1). Theorem M’s $\Delta N = -1$ edge case is general—any finite scan window produces one—but the position 402 nm is a property of the reconstruction, not of the defect. Figure 2(b) draws this rung in gray.

B. Worked example at 120 GPa

Table III and Fig. 2 give the ladder that the reconstructed kernel produces at 120 GPa, enumerated over $1 < I/I_c \leq 6$. Taking the zero-phonon height at face value, the multiplicity runs $N = 2 \rightarrow 4 \rightarrow 6 \rightarrow 4 \rightarrow 3 \rightarrow 5 \rightarrow 3$, of which the fourth step is the window-censoring event discussed below rather than a critical point of A . All six predicted step positions are reproduced to better than 0.01%, which checks the level-set enumeration against Eq. (15) rather than the values of a . The enumeration is truncated, not complete: the count of Eq. (14) requires a third interior minimum at the foot of the zero-phonon spike, and the red edge of the window supplies a further $\Delta N = -1$, both of which fall above $I/I_c = 6$. The ladder continues.

The zero-phonon rung is the least certain in the table, and it is the one that decides where the ladder starts. The first step, at $I/I_c = 1.44$, is generated by the zero-phonon line (ZPL) maximum at 514.5 nm. This rung, alone among the six, depends on a zero-phonon *height* rather than a position. Check K3 of Appendix B 2 compares the kernel’s own zero-phonon area with the published Debye-Waller factor and finds it low by 11%, by the same amount at every pressure. Correcting $a = 0.694$ by that constant gives $a \simeq 0.780$ and moves the rung

from $I/I_c = 1.44$ to 1.28. The defensible statement is therefore that the ZPL generates a rung somewhere in $1.28 \leq I/I_c \leq 1.44$, and that its optical-limit sensitivity penalty lies between 1.13 and 1.20. Neither end of that range is a bound: K3 measures an offset in *area*, while check K2 finds the zero-phonon *width* in this reconstruction to be resolution limited, and an area ratio transfers to a height only when the width is right. The range indicates a scale, not a limit.

That correction displaces one row of the ladder and does not reorder it. The critical values of a are 0.694 at the zero-phonon maximum and then 0.660, 0.658, 0.536, 0.277 and 0.233; moving the first to 0.780 carries it further from the rest rather than past any of them, so the zero-phonon step remains the lowest rung and the sequence $N = 2 \rightarrow 4 \rightarrow 6 \rightarrow 4 \rightarrow 3 \rightarrow 5 \rightarrow 3$ holds at both ends of the range. What the uncertainty moves is where the ladder starts, not what it is. Test T5 locates that first rung, and does so without absolute calibration.

What survives that uncertainty is the qualitative point, and it is the one that matters for experimental design: a candidate-line list that omits the ZPL cannot see this rung at any power. We are not aware of a candidate-line list for high-pressure work that includes the zero-phonon line, whose *photoluminescence* penalty there is unremarkable. Its sensitivity penalty is 1.13 at the optimistic end of the range above and 1.20 at the pessimistic end, where it is indistinguishable from the 1.21 of the commercial 473 nm line. On either reading it belongs on the list.

One plateau must not be claimed. The plateau between the two rungs near 475 nm spans only $\times 1.003$ in power. No realistic power resolution reaches it, and we declare it untestable rather than report it. It survives the zero-phonon uncertainty, since the pair of rungs that makes it narrow does not involve the zero-phonon line and keeps its label $N = 6$ at both ends of the range. The resolvable plateaux within the enumerated range are, at face value, the initial $N = 2$ plateau of width $\times 1.44$, followed by $N = 4, 4, 3$ and 5 of width $\times 1.05, \times 1.23, \times 1.93$ and $\times 1.19$. Correcting the zero-phonon height narrows the initial plateau to $\times 1.28$ and widens the first $N = 4$ plateau to $\times 1.18$; the other three are untouched, since no rung bounding them involves the zero-phonon line. The widest plateau is the same on either reading—the $N = 3$ plateau ending at the 500.2 nm rung, at $\times 1.93$ —and every resolvable plateau is wide enough to survive the correction.

The blue edge is a property of the figure, not of the defect. The step at $I/I_c = 1.86$ is generated by the edge of the analysis window at 402 nm, which is where the published spectrum stops rather than where the absorption band ends (Appendix B). It is counted as $\Delta N = -1$ for that reason.

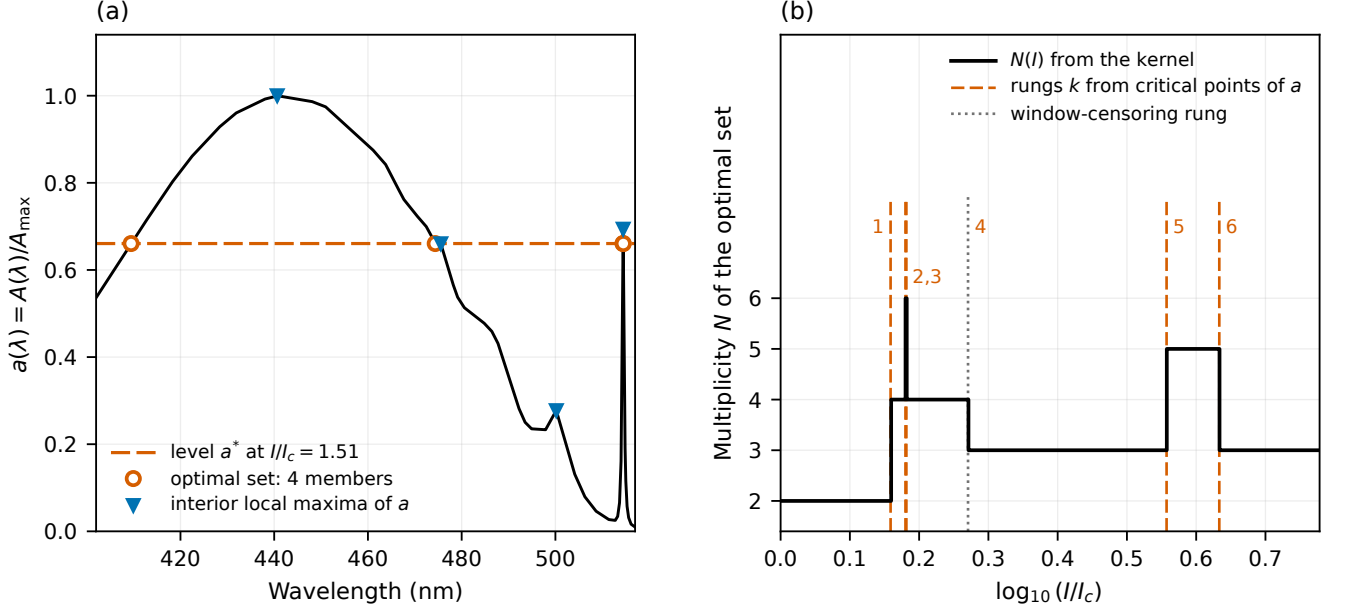


FIG. 2. The level set of Theorem M and its multiplicity ladder, for the reconstructed kernel at 120 GPa. (a) The normalized fixed-power figure of merit $a(\lambda) = A(\lambda)/A_{\max}$ with $A = \lambda\sigma_{\text{abs}}$, dimensionless, against wavelength in nm over the analysis window 402–517 nm. Triangles mark the four interior maxima (Table I); the dashed horizontal is one level I_c/I , and the circles where it cuts the curve are the members of the optimal set $\Lambda_\eta(I)$ of Eq. (13), which are exactly degenerate in η . (b) The resulting multiplicity $N = |\Lambda_\eta|$, dimensionless, against $\log_{10}(I/I_c)$. Steps occur at the powers predicted by Eq. (15) from the critical values alone; the dashed verticals are those predicted positions, and the numeral above each is the rung index k of Table III, which gives the power ratio, the generating wavelength and the step for each; rungs 2 and 3 lie 0.3% apart in power and share one numeral. Rung 4 is drawn in gray because it is generated by the edge of the analysis window rather than by a critical point of a . The $N = 6$ plateau is only $\times 1.003$ wide in power and is declared unstable. Both panels take the zero-phonon height at face value; correcting it against the published Debye-Waller factor (Appendix B 2, K3) moves that first step from $I/I_c = 1.44$ to 1.28 and leaves every multiplicity unchanged.

C. The ladder is reachable

The ladder does not require exotic powers. Reference [26] reports closed forms for R , C and $\Delta\nu$ against optical pumping power, fitted to single-NV measurements; Sec. III A converts them into $E = 3$ with $n = 2$ and $\rho^* = 1/5$, from which the sensitivity-optimal power is $\varsigma^* = 0.072$ in units of the saturation power, for the representative microwave drive $\Omega_R = 3 \times 10^6 \text{ rad s}^{-1}$. The lowest rung of the 120 GPa example then sits at

$$\varsigma = 1.44 \times 0.072 = 0.10, \quad (18)$$

that is, at about a tenth of the saturation power—*below* where many experiments routinely operate rather than above it.

Two caveats attach to that figure, and neither removes the conclusion. ς^* scales with the microwave drive (Sec. III A), so across the five settings of Ref. [26] it runs from 0.025 to 0.140. The zero-phonon uncertainty enters only weakly: the first rung moves between $I/I_c = 1.28$ and 1.44 and remains the lowest either way. Combining both, the first rung falls between 3% and 20% of the saturation power. Sweeping to ten times saturation reaches $I/I_c \leq 139$ at this drive, well beyond the highest rung

enumerated here. The ladder lies inside the accessible range by a comfortable margin.

D. The ladder is invariant across the degeneracy

The gauge degeneracy of Sec. III B, which obstructs mechanism attribution, protects the ladder. Theorem M's transition powers depend on the response only through Γ_p^* , and expressing them as I_k/I_c removes Γ_p^* entirely. Hence the ladder is invariant across the whole gauge plane. We check the collapse numerically rather than assume it: giving the five models of Fig. 1(a) five different half-scales Γ gives them five different Γ_p^* , and hence five sets of *absolute* probe intensities spanning a factor 16. Probing $N(I)$ at six points of each set—at $I/I_c = 1.05, 1.45, 1.55, 2.0, 3.7$ and 4.5 in each model's own units—returns the same multiplicities in all five cases, namely 2, 4, 4, 3, 5, 3. It returns the same six at both ends of the zero-phonon range of Sec. IV B, because that range moves the first rung between 1.28 and 1.44 and the first probe point sits below both. The probe deliberately steps over the unresolvable narrow window between the two rungs near 475 nm.

The ladder can therefore be predicted and tested *without knowing which mechanism operates*. Non-identifiability is an obstacle for mechanism assignment and a robustness property for the central prediction of this work.

V. PRESSURE-DRIVEN BRANCH EXCHANGE

Theorem M asks what power does to the optimum at one pressure. This section asks the orthogonal question, and finds an effect the ladder cannot produce.

An absorption spectrum built on a Franck-Condon progression carries two structurally distinct branches: the zero-phonon line and the phonon sideband. Their weights scale differently with the Huang-Rhys factor S , the ZPL weight collapsing as e^{-S} while the sideband weight saturates as $1 - e^{-S}$.

Theorem X (laser-dominated regime). Let $A_{\text{ZPL}}(P)$ and $A_{\text{SB}}(P)$ be the fixed-power figures of merit of the two branches, and suppose the excitation is laser dominated over the pressure range considered, $W_{\text{laser}} \geq \Gamma_{\text{ZPL}}(P)$, so that $W_{\text{eff}} = W_{\text{laser}}$ is independent of P . Let $D(P; W_{\text{eff}}) = \ln[A_{\text{SB}}/A_{\text{ZPL}}] = \ln[r(P)W_{\text{eff}}]$ with r given by Eq. (21). If r is strictly monotone in P , then D has at most one zero; and where D changes sign the global optimum exchanges branch, the optimal wavelength jumping from $\lambda_{\text{ZPL}}(P^*)$ to $\lambda_{\text{SB}}(P^*)$ taking no intermediate value.

Proof. The two branch maxima are isolated, and between them A is strictly smaller than at either, so at every pressure the global maximizer of A lies in the two-point set $\{\lambda_{\text{ZPL}}(P), \lambda_{\text{SB}}(P)\}$. It is λ_{ZPL} where $D < 0$ and λ_{SB} where $D > 0$, by the definition of D . A map into a two-point set cannot take a value between the two points, so at a sign change of D the optimum moves by the full separation $\lambda_{\text{SB}}(P^*) - \lambda_{\text{ZPL}}(P^*)$ rather than drifting through it. Monotone r gives D at most one zero, hence at most one such jump. ■

The regime hypothesis is what carries the monotonicity, and it is worth being explicit about why. Once W_{laser} falls below $\Gamma_{\text{ZPL}}(P)$, Eq. (20) pins W_{eff} to the line itself, D becomes $\ln[r(P)\Gamma_{\text{ZPL}}(P)]$, and monotone r no longer bounds the number of zeros: that now requires the *product* $r\Gamma_{\text{ZPL}}$ to be monotone, which the reconstruction cannot decide, because check K2 of Appendix B2 shows that Γ_{ZPL} is not readable from it. Restricting the theorem is therefore not a caution but the condition under which its proof runs.

Corollary X' (line-limited regime). In the complementary regime $W_{\text{laser}} < \Gamma_{\text{ZPL}}(P)$, D is independent of W_{laser} , and so is any crossing it has. Hence P^* does not move

as the laser is narrowed further: the measured curve $P^*(W_{\text{laser}})$ falls while the excitation is laser dominated and is flat below, and the knee between the two regimes sits at $W_{\text{laser}} = \Gamma_{\text{ZPL}}(P^*)$.

Corollary X' is an experimental extension rather than a second theorem: it inherits Theorem X's two-branch structure but not its monotonicity, and it predicts the *shape* of a curve rather than the existence of a crossing. Its value is that the knee measures a quantity nothing else here can reach.

The theorem is stated in the bandwidth-resolved form because that is the form this source can support. Writing the Franck-Condon pair as $A_{\text{ZPL}} \sim e^{-S}/\Gamma_{\text{ZPL}}$ and $A_{\text{SB}} \sim (1 - e^{-S})/\Gamma_{\text{SB}}$ gives

$$\frac{dD}{dP} = \frac{1}{1 - e^{-S}} \frac{dS}{dP} + \frac{d}{dP} \ln \frac{\Gamma_{\text{ZPL}}}{\Gamma_{\text{SB}}}, \quad (19)$$

whose first term is positive whenever compression strengthens the electron-phonon coupling, since $1/(1 - e^{-S}) \geq 1$. That would be sufficient for monotonicity if the ZPL also broadened at least as fast as the sideband, but Appendix B2 shows the zero-phonon width is not readable from this source at all, so we do not use it. That is precisely why Theorem X is stated in the laser-dominated regime: there $W_{\text{eff}} = W_{\text{laser}}$ is independent of pressure, Γ_{ZPL} leaves the problem, and monotonicity rests on $r(P)$ alone. Section VB establishes it there, from published data. In the line-limited regime of Corollary X' the width does not leave the problem, and the missing Γ_{ZPL} is what stops us asserting monotonicity there.

Both antecedents are generic for a color center under compression, so what varies between materials is *where* P^* falls inside the accessible range rather than *whether* an exchange occurs. That is the general condition the framework offers to high-pressure optical sensing: a once-only exchange, generic rather than an accident of one host.

The jump is large enough to matter to an experiment. In the reconstructed kernel the two branch maxima are separated by 68–78 nm at every published pressure—at 120 GPa they sit at 514.5 nm and 440.6 nm, 73.9 nm apart—so wherever P^* falls in the accessible range, the recommended excitation line moves by roughly 70 nm at that pressure instead of drifting toward it. No single line is correct on both sides.

A. The drivers, from published data

The electron-phonon driver is read from the tabulated Huang-Rhys and Debye-Waller factors of Ref. [24] itself, independently of our reconstruction of its spectra. The absorption Huang-Rhys factor rises from 3.081 to 4.613 over 0–120 GPa, an increase of 50%, monotone, at $dS_{\text{abs}}/dP = 1.27 \times 10^{-2} \text{ GPa}^{-1}$. The absorption Debye-Waller factor falls from 2.180×10^{-2} to 3.63×10^{-3} ,

a factor 6.01, monotone; both endpoints are the values Ref. [24] states in its own text, and it describes the fall as more than fivefold. The zero-phonon weight therefore collapses in the published data more steeply than a single effective mode gives: e^{-S} with the tabulated S would produce only a factor 4.63, so the published Debye-Waller factor is multi-mode, though by a margin of 1.30 rather than a wide one.

The second term of Eq. (19), the relative broadening of the two branches, is *not* confirmed and cannot be from this source: the apparent zero-phonon width in the published spectra is resolution limited (Appendix B 2, K2). Section V B therefore works in the laser-dominated regime, where the effective bandwidth replaces that width, and establishes the monotonicity of the ratio empirically instead.

Three checks support the two-branch identification. First, the two branches appear as independent local maxima of the *raw extracted samples*—not of any interpolation—at all seven published pressures. The zero-phonon branch at ambient pressure lands at 637.1 nm, i.e. 1.946 eV, which is the NV⁻ zero-phonon line and is a value fitted nowhere in the pipeline. And the two branches carry different pressure coefficients, 3.8 ± 0.2 against 5.1 ± 0.2 meV GPa⁻¹ from the seven published positions, so they are not one feature counted twice; correspondingly the Franck-Condon displacement $Sh\omega$ grows from 0.232–0.404 eV, an increase of 74%. Both coefficients come from one reconstruction of one source; the independent high-pressure optical measurements of Ref. [35] offer the external comparison that would make them harder to attribute to the reconstruction, and we have not made it. The three checks together identify the two branches as distinct features of the source rather than one feature resolved twice.

B. The exchange pressure depends on the excitation bandwidth

The two branches cannot simply have their peak heights compared, because they are different kinds of object. The sideband enters through a lineshape *density* (per unit energy), while the ZPL enters through a *dimensionless weight* that must be divided by whatever bandwidth samples it. That sampling width is not the laser linewidth alone. The absorbed zero-phonon rate is set by the overlap of the laser spectral profile with the intrinsic zero-phonon lineshape, so it grows as the laser is narrowed only until the laser is narrower than the line itself, after which it saturates. At the level of widths,

$$W_{\text{eff}} \simeq \max(W_{\text{laser}}, \Gamma_{\text{ZPL}}(P)), \quad (20)$$

and it is W_{eff} , not W_{laser} , that enters. Writing

$$r(P) = \frac{\lambda_{\text{SB}} \sigma_{\text{SB}}}{\lambda_{\text{ZPL}} \text{DWF}_{\text{abs}}} \quad [\text{eV}^{-1}], \quad \frac{A_{\text{SB}}}{A_{\text{ZPL}}} = r(P) W_{\text{eff}}, \quad (21)$$

TABLE IV. The drivers of Theorem X. DWF, Debye-Waller factor; SB, phonon sideband. Taken from the published theory of Ref. [24] [its panels (b) and (c)] rather than from the reconstructed spectra. $r(P)$ is the branch ratio per unit excitation bandwidth, Eq. (21); $1/r$ is the bandwidth below which the ZPL branch is competitive.

P (GPa)	0	20	40	60	80	100	120
S_{abs}	3.081	3.368	3.648	3.908	4.152	4.375	4.613
$\text{DWF}_{\text{abs}} (10^{-2})$	2.180	1.565	1.127	0.835	0.620	0.479	0.363
$r (\text{eV}^{-1})$	97.1	121.9	162.1	208.0	267.1	329.1	418.4
$1/r (\text{meV})$	10.30	8.20	6.17	4.81	3.74	3.04	2.39

the exchange occurs where $r(P) = 1/W_{\text{eff}}$. Table IV shows that r is monotone increasing, by a factor 4.31 over the published range, so at most one crossing survives and Theorem X’s monotonicity antecedent holds across that range. What changes is that P^* is no longer a single number, and that for bandwidths outside 2.4–10.3 meV no crossing falls inside the accessible range at all.

Figure 3(b) gives the dependence, at fixed incident optical power: P^* falls from 117 GPa at $W_{\text{eff}} = 2.5$ meV through 101 GPa at 3 meV to 13 GPa at 9 meV, each value carrying about 2 GPa from the 3% uncertainty on r ; at fixed incident photon flux each moves by about 10 GPa. A laser linewidth chosen anywhere in the commercial range therefore moves the exchange across the whole accessible pressure axis. Equivalently, the ZPL competes at all only for an excitation bandwidth below $1/r(P)$, which narrows from 10.3 meV at ambient pressure to 2.4 meV at 120 GPa.

The bandwidth dependence is a sharply testable prediction, and Eq. (20) sharpens it further. By Theorem X the exchange pressure moves monotonically with the laser linewidth while the excitation is laser dominated, and by Corollary X’ it stops moving below that. A measured $P^*(W_{\text{laser}})$ curve should therefore fall and then flatten, and the *knee* between the two regimes locates Γ_{ZPL} at that pressure—a quantity check K2 of Appendix B 2 shows cannot be read from the reconstruction at all. No other mechanism in this framework produces either the falling segment or the knee; a fixed-wavelength ranking crossover has neither. It also raises the value of test T11 of Sec. VI, which probes the electron-phonon driver without touching the bandwidth question at all.

Reporting rule. A value of P^* must never be quoted without stating both the excitation bandwidth and the fixed-power versus fixed-flux convention.

C. What this is not

Two nearby phenomena would, if conflated with branch exchange, make a generic effect look like an accident.

It is not a fixed-wavelength ranking crossover. In the same kernel the sensitivity ranking of 457 nm and 532 nm

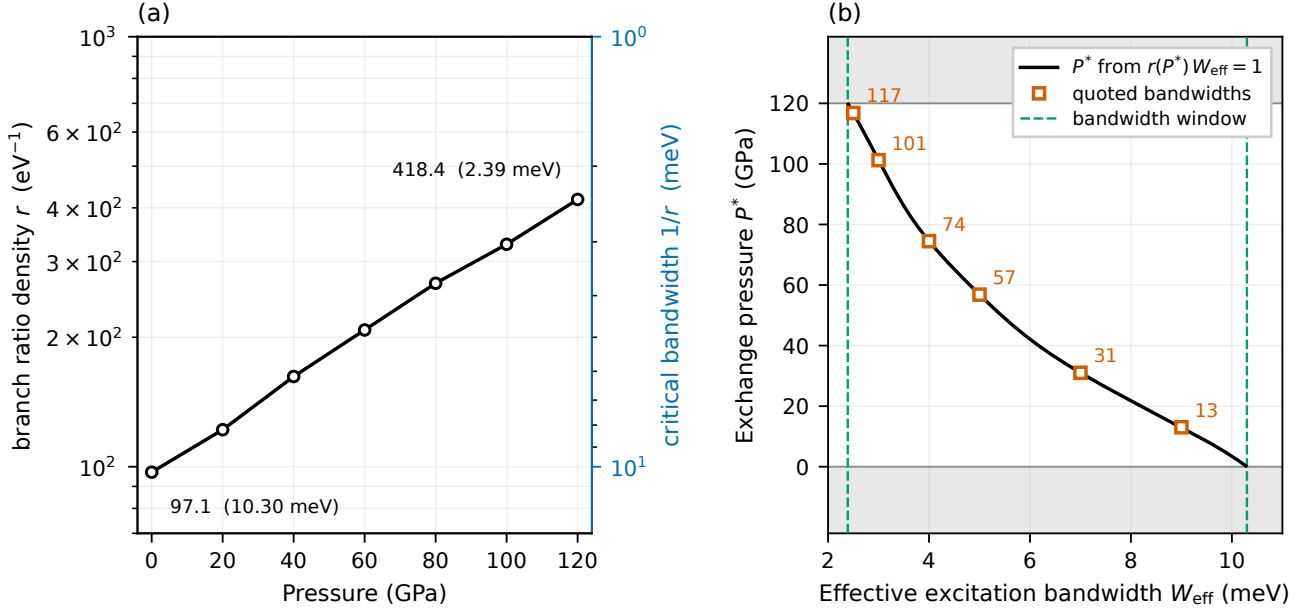


FIG. 3. The branch exchange of Theorem X. SB, phonon sideband; ZPL, zero-phonon line. (a) The branch ratio per unit excitation bandwidth, $r(P) = \lambda_{\text{SB}} \sigma_{\text{abs}}^{\text{SB}} / (\lambda_{\text{ZPL}} \text{DWF}_{\text{abs}})$ of Eq. (21), in eV^{-1} , against pressure in GPa on a log ordinate. It rises monotonically by a factor 4.3 over 0–120 GPa, which is what makes the crossing of Theorem X unique where it occurs. The right axis is the critical bandwidth $1/r$ in meV, below which the zero-phonon branch is competitive; the annotated endpoints are 10.30 meV and 2.39 meV. (b) The exchange pressure P^* , in GPa, against the effective excitation bandwidth W_{eff} of Eq. (20), in meV, at fixed power. Square markers are six bandwidths spanning the accessible window, 2.5 meV, 3.0 meV, 4.0 meV, 5.0 meV, 7.0 meV and 9.0 meV; the numeral beside each is the exchange pressure it produces, and P^* carries ± 2 GPa. The two dashed verticals bound that window and the shaded bands are outside the published pressure range, which no crossing enters for W_{eff} outside roughly 2.4–10.3 meV. Sweeping the laser linewidth moves P^* monotonically, and no other mechanism in this framework does so.

reverses near 51 GPa, which is the geometry of a single band sweeping past the midpoint of two fixed probes and requires no branch structure at all. That pressure corresponds to a bandwidth of about 5 meV and lies inside the branch-exchange band of Fig. 3(b), so the two cannot be told apart by where they fall. The discriminant is mechanism, and it is test T9: a ranking crossover involves one peak, an exchange requires *both* branches on either side of P^* .

It is not visible to a single-mode envelope. A single-effective-mode Franck-Condon envelope has exactly one interior local maximum at every pressure between 0 and 120 GPa, so it has no branches to exchange and any crossover it shows is the fixed-wavelength kind, at a pressure of its own. The reliable discriminant is to count the interior local maxima of $A(\lambda)$ in each model—the same count that decides whether Theorem M has a ladder at all—rather than to compare two pressures and read a coincidence as agreement.

D. The (P, I) plane

Theorems M and X produce degeneracies of different origin—one along the power axis, one along the pressure

axis—and they stack.

- *Below P^** , the zero-phonon branch carries the global optimum, so the first members of $\Lambda_\eta(I)$ lie on it. The level set is defined over the whole window, so as the level falls it crosses critical values on the sideband as well, and those wavelengths join the optimal set.
- *At P^** , the low-power optimum is already twofold, one member on each branch, before any power-induced structure appears. Raising the power then adds rungs drawn from both. This is the point of richest structure.
- *Above P^** , the sideband branch carries the global optimum and supplies the first members; the zero-phonon branch enters the optimal set once the level falls to its peak.

The ladder therefore mixes branches rather than being confined to one, and the worked example shows it doing so. At 120 GPa, which lies at or above P^* for every bandwidth in Fig. 3(b), one rung of Table III is generated by the zero-phonon maximum, four are sideband features and one is the window edge—and on the face-value branch the zero-phonon rung is the lowest of them.

What the pressure fixes is which branch carries the *global* optimum, and hence the order in which the two branches enter the ladder; it does not assign the ladder to a branch.

Theorem M is thus the structure along the I axis and Theorem X the structure along the P axis; together they give the phase diagram of the optimal operating point.

The practical discriminant between them is that *Theorem X's degeneracy exists at zero power while Theorem M's requires $I > I_c$* . At P^* the optimum is already twofold in the low-excitation limit, so the two can be separated by measuring at low power and varying pressure, before any power-induced multiplicity appears. Test T10 is that separation.

VI. PRE-SPECIFIED TESTS

Each of the three theorems makes predictions that can fail, and the value of stating them in advance is that none can be adjusted after the data are in. Table V collects the eleven tests they license, grouped by the claim each one probes.

T4 converts a single-wavelength measurement into a prediction about a multi-wavelength scan. On synthetic data—fourteen powers at one wavelength, 2% multiplicative noise— Γ_p^* is recovered to 2.7% in a single trial, and across 200 trials the 16–84% interval is about $\pm 2\%$. Combined with T5, which supplies the a_k from a low-power scan, the whole ladder is predicted before any wavelength-resolved sensitivity measurement is made.

T6 is a null test of arbitrary precision. Under (M) the degeneracy among the members of Λ_η is exact, so any measured deviation is itself a measurement of how far (M) fails.

T11 is the cheapest. The branch separation is a difference of two peak positions, so it needs neither absolute nor intensity calibration. One caution: the separation is $S\hbar\omega$, and compression hardens $\hbar\omega$ as well as raising S . Isolating S requires the stationarity condition of the Franck-Condon envelope, $\psi(p^* + 1) = \ln S_{\text{abs}}$ with ψ the digamma function and p^* the observed sideband maximum in units of $\hbar\omega$; the continuum shortcut $p^* = S_{\text{abs}}$ is wrong by about half a phonon over the relevant range. If the separation does not grow with pressure, the electron-phonon driver of Theorem X fails and no exchange is predicted.

We also record the falsification conditions explicitly. If T1 collapses onto a single curve and $\lambda_\eta = \lambda_{\text{PL}}$ within 3 nm at every power, then no divergence exists, the conventional practice of maximizing photoluminescence is correct, and the central proposition of this work fails. If the predicted rungs I_k/I_c carry no steps within the power resolution, Theorem M fails—either (M) does not hold or Φ has more than one interior maximum. If $N(I)$ moves continuously rather than in steps, η is not a function of Γ_p alone and (M) is violated. If $\lambda_{\text{PL}}(P)$ moves continuously through P^* , or if the laser linewidth can be varied without moving P^* , Theorem X and its bandwidth

corollary fail respectively.

VII. DISCUSSION

Two of the framework's three antecedents have already been measured by others, one has not, and the structures it reports have partial precedents in sensing that are worth separating from it precisely. What follows takes those in turn, then asks whether charge conversion alone could produce the same effect, what the framework settles that was previously open, why the effect belongs to high pressure, and where its limits lie.

A. Relation to what has already been measured

Two of the three antecedents this framework needs have already been measured. The interior maximum of Φ —the condition $En > 1$ without which there is no level set, no degeneracy and no ladder—is the content of the closed forms Ref. [26] fitted to measurement in 2011, robust across all the microwave settings that work reports (Sec. III A). The wavelength dependence of the ODMR contrast, $\ell_G \neq 0$, has been measured at ambient pressure by Ref. [23]. Neither is assumed here.

What that work leaves open is what remains. It optimizes the contrast rather than the sensitivity, so the linewidth—which enters η with the same power as the contrast—is not recorded; and it re-optimizes the excitation power at each wavelength, which is exactly the operation that destroys the fixed-power comparison Theorem M is built on, collapsing a two-dimensional optimization over (λ, I) onto a section through it. The multiplicity of the optimal set, its degeneracy, the ladder and the gauge structure are therefore untouched. One measurement would connect that work to this one: contrast and linewidth recorded separately at a common power.

B. Related notions of multiplicity and of exchange

Neither of the two structures this paper reports is without precedent in sensing, and it is worth saying precisely where each departs from what exists.

Multiplicity first. Ma *et al.* established that an absorption-based sensor is designed around a *set* of transitions rather than one, the set being chosen to condition the inverse problem that maps measured absorbances onto a nonuniform temperature field [36]. Sun *et al.* treated the several resonances of a nonlinear photonic gyroscope as a single joint resource and maximized the Fisher information of the whole cavity [37]. In the first the multiplicity is selected by the experimenter and its members carry different information; in the second the resonances cooperate, and the figure of merit is a functional of all of them together. Here the members of $\Lambda_\eta(I)$ are alternatives, each individually optimal and mutually

TABLE V. Pre-specified tests. Each entry states what an outcome is consistent with and what it falsifies. Agreement of finitely many observables at finite precision constrains (M) and does not establish it: a photon-energy-specific channel that happened to track $A(\lambda)$ over the measured range would pass every forward test. (M) is therefore strongly falsifiable and not uniquely provable from this set. T1–T4 probe the mediation hypothesis (M) and the coincidence condition; T5–T7 test Theorem M and Theorem G; T8–T11 test Theorem X. None requires absolute calibration: every entry is a ratio, a step position, or an equality.

test	procedure	discriminates
T1 collapse	plot η against the measured R for all (λ, I) , discarding λ	collapse is consistent with (M); the scatter measures ℓ_G
T2 power closure	measure $\lambda_\eta(I) - \lambda_{PL}(I)$ over a power sweep	closes as $I \rightarrow 0$: consistent with intensity mediation; a persistent offset falsifies (M)
T3 degeneracy	compare η at the members of a doublet, searching for equal η rather than symmetric placement	equality within the measurement error supports mediation by the pump rate; an imbalance measures ℓ_G and falsifies (M)
T4 single-line prediction	from one power sweep at a single wavelength, extract the response scales and hence Γ_p^*	predicts I_c and the ladder <i>without any wavelength scan</i>
T5 ladder	read the critical values a_k off a low-power $R(\lambda)$ scan at fixed incident optical power; predict $I_k/I_c = 1/a_k$; verify the steps	steps at the predicted powers support Theorem M; steps elsewhere, or none, falsify it
T6 universal degeneracy	compare η across <i>all</i> N members of a multiplet	equality at the noise floor bounds any violation; the deviation measures it
T7 exponent closure	measure $C(I)$, $R(I)$, $\Delta\nu(I)$ separately; form $E = 2c + s + 2w$ and the pump nonlinearity n ; predict $\rho^* = 1/(En - 1)$	agreement is consistent with a power-law response; disagreement falsifies the assumed functional form
T8 exchange	track $\lambda_{PL}(P)$ at low power through P^*	a discontinuous jump supports Theorem X; continuous motion falsifies it
T9 branch coexistence	check that both peaks are present on either side of P^*	both present is consistent with exchange; only one indicates branch disappearance, a different mechanism
T10 zero-power degeneracy	at $P \approx P^*$ compare η at the two wavelengths in the low-excitation limit	equal at zero power indicates Theorem X; a power threshold indicates Theorem M
T11 driver	measure the branch separation $S\hbar\omega$ against pressure, and invert $\psi(p^* + 1) = \ln S$ to isolate S	monotone increase of S establishes the antecedent of Theorem X; a constant S predicts no exchange

degenerate in η , nothing selects them, and it is their *number* rather than their combination that is the observable.

Exchange second. Shaji Rebeirro *et al.* demonstrated microwave-free wide-field NV magnetometry at the ground-state level anticrossing, one member of a family of operating features that also contains the excited-state anticrossing and the zero-field crossings [38]. Ronchi *et al.* showed that the quantum Fisher information of a probe at an avoided crossing is maximized by coherent evolution in one region of the offset-against-available-time plane and by repeated projective measurement in the other, so that the optimal strategy exchanges across a boundary in that plane [39]. The branches of the first are branches of the static bias field and the operator selects among them by setting that field; what exchanges in the second is the measurement protocol applied to one transition. Here neither the branch nor the protocol is chosen: which physical transition carries the optimum is fixed by the pressure, under a protocol held constant, and the exchange is a discontinuity in an external control variable rather than a decision.

C. Charge conversion alone cannot split the optima

It is natural to expect the NV charge state to be what separates the two optima, since it is the most wavelength-sensitive quantity in the problem. It cannot be, and the reason is structural. In the frozen charge model both ionization from the excited state and recombination through excitation of NV^0 scale with the same absorption cross section, so the steady-state NV^- fraction

$$f_- = \frac{r_0\sigma + r_{bg}}{(r_0 + a_{es})\sigma + r_{bg}} \xrightarrow{\sigma \gg r_{bg}/r_0} \frac{r_0}{r_0 + a_{es}} \quad (22)$$

is a constant wherever the cross section is appreciable. Here $\sigma \equiv \sigma_{abs}$, r_0 is the recombination-rate coefficient for excitation of NV^0 , a_{es} the excited-state ionization coefficient, and r_{bg} a cross-section-independent background recombination rate. More generally, f_- is mediated by Γ_p in the sense of (M), so it enters the three cases of Sec. II E through A alone and cannot itself be the source of a separation between λ_η and λ_{PL} .

The corollary is sharp and identifies the only charge-related channel that survives: charge conversion can

move the sensitivity optimum *only* if $\sigma_{\text{abs}}^{\text{NV}^0}$ has a spectral shape different from that of $\sigma_{\text{abs}}^{\text{NV}^-}$, so that the two rates cease to be proportional. That is a violation of (M), not an instance of it. It is also the channel that the 575 nm feature of Ref. [23]—the NV^0 zero-phonon line—is evidence for, and the channel that Ref. [22] probes. The optimum excitation wavelength reported there for single centers at ambient pressure is an optimum of the *rate*, arising from the recharging balance; if that balance carries wavelength structure of its own, it is structure of exactly this kind. What Eq. (22) shows is that the frozen-charge part of the balance cannot supply it.

D. An open question this framework answers

Reference [21], reviewing absorption-based readout, records that the anomalous contrast reveals a strong wavelength *and* power dependence which suggests that green absorption readout is hindered by an unknown effect, that the wavelength and power dependence of that effect suggests NV^0/NV^- charge dynamics could play a role, and that further investigation might reveal presently unknown NV dynamics [21, 40].

That is a description of the (λ, I) plane this paper organizes, and Sec. VII C answers it: charge conversion alone cannot produce the effect, and the only channel that can is a difference in spectral *shape* between the NV^0 and NV^- bands.

E. Why the effect belongs to high pressure

Theorem M is a statement about any absorbing sensor and could be tested at ambient pressure; only the values of its rungs are pressure specific. Theorem X could not: it needs $dS/dP > 0$ (Sec. I), so whether P^* falls inside the accessible range is what decides observability in a given host.

F. Limitations

Five limitations remain.

(M) is *unverified*, and is known to fail at discrete resonances. The resonances identified so far [23] lie redward of 500 nm, outside the blue window that carries the high-pressure optimum, but they are ambient-pressure measurements and the levels shift under compression. We therefore present (M) as the object of test T6, and claim only that it can be tested in the blue window to arbitrary precision.

Every number is conditional on one reconstruction, with node spacing of 3–10 nm in the blue window and piecewise-linear interpolation between nodes; the zero-phonon critical *value* carries in addition the constant 11% offset against the published Debye-Waller factor measured by check K3 of Appendix B 2, which places the first

rung to about $\pm 6\%$ in power. The theorems are what we freeze. In an experiment the critical values a_k are read from the measured spectrum, not taken from Table III.

Theorem M assumes a unique interior maximum of Φ . If there are several, the optimal set is a union of level sets and the ladder is correspondingly more complex. For the power-law responses of Eq. (7) with $En > 1$ the maximum is unique, so test T7 also constrains this premise.

Theorem X assumes two branches. A spectrum with three or more structurally distinct branches would exchange more than once, giving a sequence of exchanges rather than a single one—the pressure-axis analogue of the multiplicity ladder.

ZPL excitation is bandwidth sensitive in a way the other lines are not, and the sensitivity is shared by Theorems M and X. If the laser linewidth exceeds the ZPL width, the effective $A(\lambda_{\text{ZPL}})$ falls and the lowest rung of the ladder moves to higher power. Any experiment including the ZPL among its candidate lines must record the laser linewidth, for the same reason that P^* must be quoted with a bandwidth.

VIII. CONCLUSIONS

Selecting the excitation wavelength of an NV magnetometer by maximizing its photoluminescence requires $C/\Delta\nu$ to be wavelength independent. That condition is testable, and it is already known to fail. Relaxing it does not displace the optimum; it changes what kind of object the optimum is.

Under the single hypothesis (M), and with no assumption about the absorption spectrum, the sensitivity optimum is a level set whose members are exactly degenerate and whose cardinality is a staircase in $\ln I$. The response enters that staircase only through $E = 2c + s + 2w$ and the pump nonlinearity n , so the mechanism cannot be recovered, within that response class, from sensitivity data at any number of powers, and the ladder is correspondingly robust to not knowing it. And because compression raises the Huang-Rhys factor, the branch carrying the global optimum is exchanged discontinuously, at a pressure the experimenter selects through the excitation bandwidth.

Two of the three antecedents have already been measured by others, at ambient pressure, and the lowest rung of the ladder falls between 4% and 21% of saturation—inside the range at which experiments already operate. Eleven pre-specified tests follow, none requiring absolute calibration; T11 is the cheapest, needing only two peak positions per pressure, and T6 is a null test of arbitrary precision for (M) itself. In that sense the framework supplies a way to measure its central hypothesis instead of assuming it.

The stakes are set by Eq. (1). Map acquisition time scales as η^2 , so the commercial 473 nm line costs a factor 1.45 in time against the optimum at 120 GPa, conditional on the reconstruction (Appendix A)—and on the far side

of P^* the correct line lies on the other branch entirely.

Appendix A: Fixed-power optical limit at 120 GPa

The results of the main text are structural. For reference we record the fixed-power optical limit of the reconstructed kernel, which is the input the theorems act on and the quantity a laser specification would be drawn from.

Assumptions: fixed incident optical power, low saturation, and wavelength-independent charge yield, ODMR contrast, linewidth and collection within the scan—that is, $\ell_G = 0$, so that by Eq. (6b) the two optima coincide and $\eta \propto a^{-1/2}$. Under exactly these assumptions the interpolant maximum is 440.6 nm, reported as approximately 441 nm because the extracted blue-side nodes are 3–10 nm apart. The 5% penalty band is 426.4–457.9 nm and the 10% band 420–465 nm. Table I gives the penalties.

Of the commercial lines conventionally proposed for high-pressure NV work, 445 nm and 457 nm fall inside the 5% band, at penalties of 1.004 and 1.045; the 473 nm line costs 21%. The ground-state photoionization threshold at 120 GPa, $\text{IP}(^3A_2) = 3.06$ eV or 405.2 nm, imposes a hard blue boundary: it is active only below 405 nm and therefore sets a wall without displacing a 440 nm optimum. It is one of the identified channels through which (M) can fail, but it fails outside the window of interest.

1. Extrapolation to 300–400 GPa: a blue wall, not a unique optimum

The preceding result is anchored at 120 GPa. It is nevertheless useful to ask what a controlled continuation would imply for the 300–400 GPa regime. We use the pressure-dependent ZPL and ground-state ionization energy in Ref. [24] as the anchor, and impose the physically motivated constraint that they are continued by the same law: the authors report that $\text{IP}(^3A_2)$ closely follows the ZPL shift. For any such continuation, the one-photon ionization boundary is

$$\lambda_{\text{ion}}(P) = \frac{hc}{\text{IP}(^3A_2, P)}, \quad hc = 1239.84 \text{ eV nm}. \quad (\text{A1})$$

Excitation at $\lambda < \lambda_{\text{ion}}$ supplies a photon above the ground-state threshold and can deplete NV^- into NV^0 ; the corresponding photoionization channels and their state dependence are established by the ab-initio analysis of Ref. [29].

Table VI gives the resulting estimates at the target pressures. The optimum range is the envelope of the internally consistent variants: saturating versus linear ZPL continuation and the two illustrative phonon-hardening factors. The 120 GPa row is deliberately not replaced: its independently reconstructed kernel optimum is the 440.6 nm value reported above. The pressure-dependent

TABLE VI. Illustrative continuation above the 120 GPa anchor. The ionization wall uses the same continuation for the ZPL and $\text{IP}(^3A_2)$. The optimum interval spans the consistent model variants; it should not be read as a confidence interval.

P (GPa)	$\text{IP}(^3A_2)$ (eV)	λ_{ion} (nm)	λ_{opt} (nm)
300	3.63	341.6	349–451
400	3.95	314.1	314–448

optical kernel itself is measured only to 120 GPa, so this envelope is an uncertainty statement, not a prediction of a unique laser line.

The practical conclusion is one-sided. Once the optimum reaches the blue wall, making the excitation wavelength still shorter cannot improve the sensing rate: it increases the photon energy beyond the ionization threshold and reduces the NV^- population that produces ODMR. In the linear, coupled continuation the wall is therefore about 342 nm at 300 GPa and 314 nm at 400 GPa. In a saturating continuation it remains near the 405 nm anchor value. These alternatives cannot be distinguished by the present data. Moreover, pressure calibration itself becomes a material systematic in this range: diamond-edge Raman scales have been demonstrated to about 370 GPa [41] and extended to approximately 0.5 TPa, with substantial differences between competing scales above 300 GPa [42]. A 300–400 GPa excitation optimization is consequently feasible as an experiment, but not yet identifiable as a unique calculation from the 120 GPa record.

2. Stress anisotropy and the status of the α correction

The excitation wavelength can also depend on the stress tensor, not only on its nominal pressure coordinate. For an NV axis along [111], we parameterize an axisymmetric tensor by

$$\sigma(P, \alpha) = P [\alpha \mathbf{1} + (1 - \alpha) \hat{\mathbf{n}}_{111} \hat{\mathbf{n}}_{111}^T], \quad \hat{\mathbf{n}}_{111} = \frac{(1, 1, 1)}{\sqrt{3}}, \quad (\text{A2})$$

where compression is positive. In the NV frame this is $\text{diag}(\alpha P, \alpha P, P)$, with mean stress $\bar{\sigma} = (1 + 2\alpha)P/3$ and axial deviatoric excess $\sigma_{\text{dev}} = (1 - \alpha)P$. Thus $\alpha = 1$ is hydrostatic, whereas smaller α represents increasing transverse stress deficit.

This distinction is important for interpreting the wavelength results. The 440.6 nm optimum reconstructed from Ref. [24] is a hydrostatic optical-kernel result: α does not enter that kernel, so the reported optimum is not an orientation-resolved prediction. The phenomenological model used for the conditional stress analysis does include α in the ZPL shift, calibrated to the ratio of the standard-culet and micropillar responses reported by Hilberer *et al.* [13]. At equal nominal axial stress their

reported values correspond approximately to $\alpha = 0.56$ for a standard flat culet and $\alpha = 0.95$ for a micropillar; the latter is close to quasihydrostatic loading. The earlier hydrostatic and uniaxial-stress analysis of Doherty *et al.* [9] supplies the physical basis for treating the ZPL as stress-sensitive.

The α correction is therefore only partial. It changes the ZPL position, but the present data do not determine the corresponding α -dependence of the Huang–Rhys factor, phonon-sideband shape, linewidth, charge-state yield, or the full absorption kernel. Consequently, the stress-corrected wavelength is a conditional estimate, not a literature-validated value of $\lambda_{\text{opt}}(P, \alpha)$. In particular, substituting $\bar{\sigma}$ for P in the hydrostatic kernel would simply assume an unmeasured mapping. The required validation is a wavelength-resolved absorption/ODMR scan at several independently reconstructed α values, with the local stress tensor measured simultaneously.

Appendix B: Domain of validity of the reconstructed kernel

The worked examples of Secs. IV and V descend from one reconstruction of published curves—the exceptions are the reachability estimate of Sec. IV and the drivers of Table IV, both taken from published theory—and nothing in the framework’s own logic can say whether that reconstruction is sound. The check has to come from the source.

1. What the cross-figure check does and does not cover

The reconstruction reproduces the independently published pressure-dependent photoluminescence yield with a pooled fractional RMS of 1.0%, correlation $r = 1.00$, and peak pressures exact at 17 and 88 GPa, across the full published 0–120 GPa axis [Fig. 4]. That is a strong check where it applies.

It probes exactly two photon energies, and both lie deep in the phonon sideband throughout their audit windows: 532 nm (2.331 eV) over 4.7–51 GPa, sitting 5.5 to 2.2 phonons above the ZPL, and 457 nm (2.713 eV) over 51–113.8 GPa, sitting 8.1 to 4.9 phonons above it. *The zero-phonon line is never sampled.* The 1.0% agreement must therefore never be cited as validating the ZPL region, which is precisely where the reconstruction is known to be defective.

2. Four internal checks

Four checks internal to the same source close that gap [Fig. 5].

K1: the blue edge of the analysis window is a figure limit. The published spectra are drawn to unit area, and

the digitized curves confirm this to 0.5% at 0–60 GPa. Above that the integral falls, to 0.914 at 120 GPa, because the band does not return to zero inside the plotted window: absorption at 3.0–3.2 eV is still 72.7% of its own peak at 120 GPa, against 0.0% at ambient pressure. The 402 nm blue edge is therefore a property of the figure, not of the defect, and the $\Delta N = -1$ step it generates in Table III is labeled an edge for that reason. Area-based quantities are underestimated at high pressure by the missing tail.

K2: the ZPL width is resolution limited. The apparent full width at half maximum of the zero-phonon spike is 2.53–2.59 meV at all seven published pressures, constant to 2.4% over 120 GPa. A physical ZPL broadens under compression—that is the second term of Eq. (19). A width that does not move is the plotting resolution. No ZPL width or broadening may be read from the reconstruction, which is why Sec. VB replaces Γ_{ZPL} by the excitation bandwidth rather than attempting to measure it.

K3: the kernel’s ZPL carries a constant offset, not a trend. Integrating the background-subtracted spike gives the kernel’s own Debye-Waller factor. It is low against the published value by 9–12% at every one of the seven pressures, with no monotone drift, and the two collapses over 0–120 GPa agree to 1.3%— $\times 6.08$ for the kernel against $\times 6.01$ for the published factor. The *shape* of the zero-phonon pressure dependence is therefore reproduced by the reconstruction; what it misses is a scale. This is an independent validation in the region the cross-figure check cannot reach, and it is why the zero-phonon critical value of Table I is quoted as a range rather than a number. Theorem X is nonetheless evaluated from the published Debye-Waller factor (Table IV) rather than from the reconstruction, because there the absolute scale is what enters.

An earlier raster extraction of the source’s panel (c) carried a constant baseline offset and put this ratio at 0.96 rising to 1.43, which was read as the reconstruction over-weighting the zero-phonon line above 40 GPa. That trend was an artifact of the extraction. The vector reading used here reproduces the two endpoint values Ref. [24] states in its own text, 2.2% and 0.36%, to better than 1%.

K4: the two published panels are internally consistent. The Debye-Waller factor is not $\exp(-S_{\text{abs}})$, the ratio growing smoothly from 2.11 to 2.73; this is expected, since S_{abs} is defined over the Jahn-Teller-inactive modes only. The residual is the Jahn-Teller-active coupling, $S_{\text{JT}} = \ln[e^{-S_{\text{abs}}}/\text{DWF}]$, which runs from 0.745 to 1.005 and gives S_{total} from 3.826 to 5.619. Two separately digitized panels yielding a smooth, monotone derived quantity is itself evidence that neither extraction is broken.

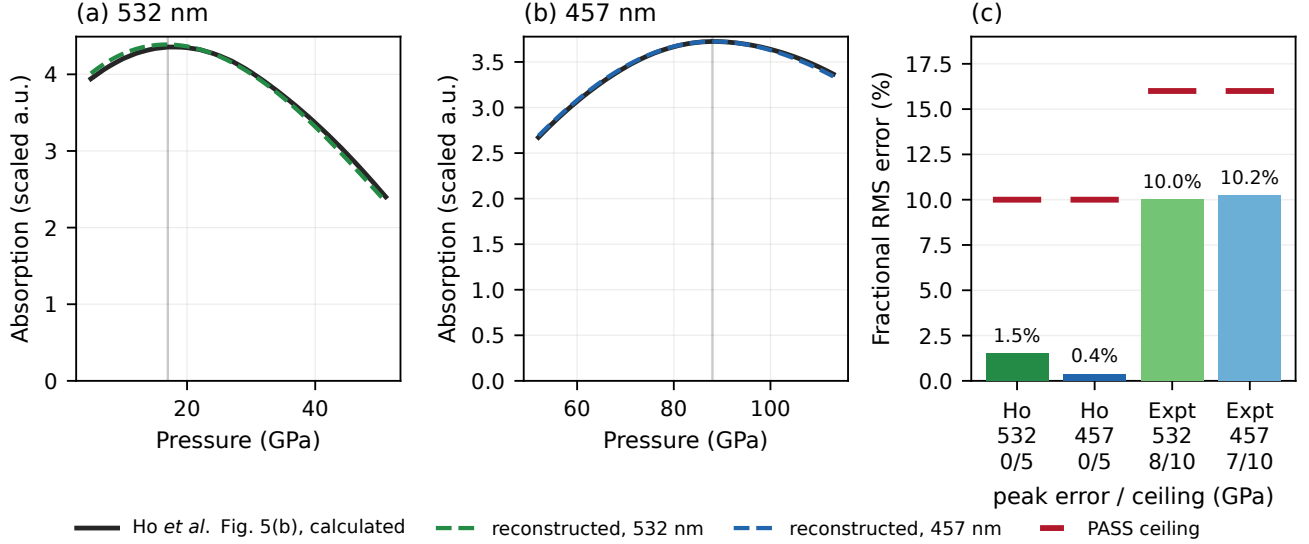


FIG. 4. The cross-figure check on the reconstructed kernel. Expt, the experimental gate; Ho, the theory gate of Ref. [24]. The absorption curve is reconstructed without reference to the independently digitized photoluminescence-yield curves, then compared with them at the two wavelengths those curves plot, (a) 532 nm and (b) 457 nm, giving fractional RMS residuals of 1.5% and 0.4% and reproducing the yield peaks at 17 and 88 GPa exactly; the vertical rule in each panel marks the published peak. The reconstruction carries one multiplicative scale and no shape freedom. (c) Those two residuals, and the two larger ones of 10.0% and 10.2% obtained against the measured yields, each against its acceptance ceiling (red dash); the abscissa labels give the peak-position error and its ceiling in GPa. The figure validates an extraction of published curves; it is not an independent first-principles calculation.

3. Two regions that must not be used

Two spectral regions of the reconstruction carry no data and must not be quoted.

Below the zero-phonon line. The lowest-energy real sample lies at 2.396 eV (517.4 nm), just below the reconstructed ZPL at 2.410 eV (514.5 nm); the only other sample below that is a single baseline point at the figure’s axis edge. 532 nm (2.331 eV) falls inside that empty interval, so the ratio of 457 nm to 532 nm sensitivity—nominally a factor 12 in the optical limit—is produced by interpolation, not by extracted data. Physically, absorption below the ZPL vanishes at $T = 0$ and is thermally activated,

$$\sigma_{\text{abs}}(E_{\gamma} < E_{\text{ZPL}}) \propto \exp\left[-\frac{E_{\text{ZPL}} - E_{\gamma}}{k_B T}\right], \quad (\text{B1})$$

with $E_{\text{ZPL}} - E_{532} = 79.5 \text{ meV}$ here, giving 4.6×10^{-2} at 300 K and 3.5×10^{-5} at 90 K. The interpolated value 6.4×10^{-3} lies between these without deriving from either. Any comparison involving 532 nm at 120 GPa is governed by hot-band absorption and temperature and must be treated as temperature dependent, or the discussion restricted to $\lambda \leq 517 \text{ nm}$ where extracted data exist.

The zero-phonon peak heights. Every ZPL spike in the source figure rises to within one unit of the top of the axis—measured tops span only 0.92 units across all seven pressures on an axis ending near 15—so the spikes are drawn as near-delta lines and the plot simply cuts them

off. Their reconstructed heights are a digitization artifact of a clipped feature. Taken with K2, the only trustworthy zero-phonon information in the source figure is its *position*. This is why Sec. V takes the ZPL weight from the published Debye-Waller factor, and why P^* is reported as a function of bandwidth rather than as a number.

These restrictions reach Sec. IV unevenly. All four critical points of Table I lie at $\lambda \leq 514.5 \text{ nm}$, inside the region where extracted data exist, so their *positions* are sound to the node spacing—0.1–0.5 nm at the zero-phonon point, 3–10 nm in the sideband. One *height* is affected, and it is the one Theorem M needs: $a(\lambda_{\text{ZPL}})$ is a zero-phonon peak height, which is why Sec. IV B carries it as a range and propagates both branches of the resulting ladder. In an experiment the point is moot, since T5 reads the a_k from the measured spectrum.

Appendix C: Reproducing the numbers

Every number in this paper is produced by the code in `code/` and pinned by the regression suite.

- `ho_spectrum_model.py`, `ho_odmr_sensitivity.py` — the reconstructed kernel and the fixed-power sensitivity.
- `report_120gpa_sensitivity.py` — the laser-line block of Table I.

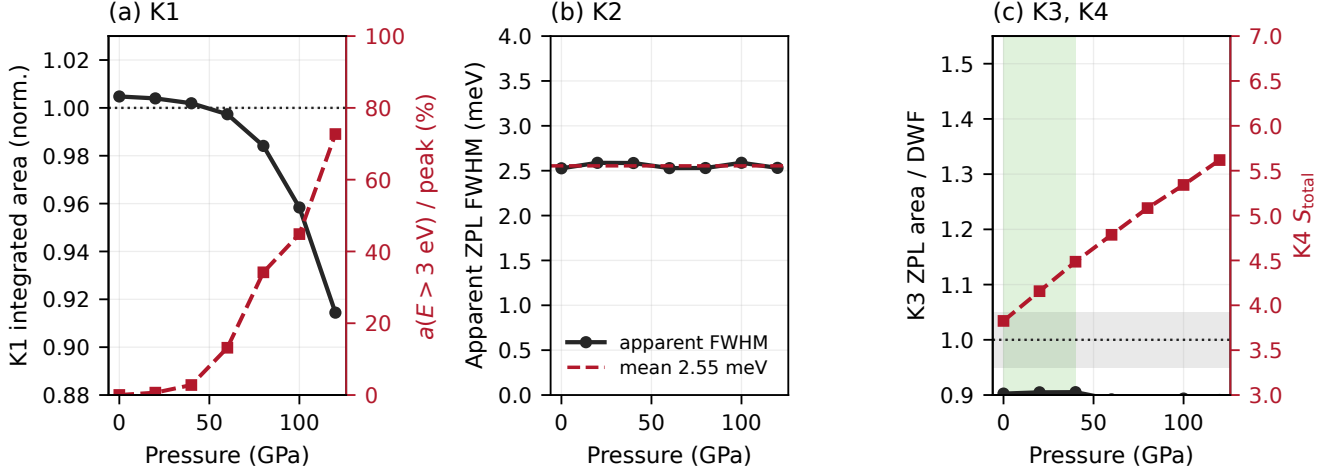


FIG. 5. The four checks internal to the source, K1 to K4. ZPL, zero-phonon line; DWF, Debye-Waller factor. In (a) and (c) each curve is drawn in the color of the axis that scales it. (a) K1, the area normalization against pressure (left, black), falling to 0.914 as the band ceases to return to zero inside the plotted window, which the weight remaining at the top of that window (right, red) tracks: 0.0% of the peak at ambient pressure and 72.7% at 120 GPa. (b) K2, the apparent zero-phonon width in meV, constant to 2.4% over the full range and therefore resolution limited rather than physical. (c) K3, the kernel's own zero-phonon area against the published Debye-Waller factor (left, black), with the gray band marking $\pm 5\%$ around parity and the green span the pressures below 40 GPa over which the kernel ZPL may be used; and K4, the total coupling $S_{\text{total}} = S_{\text{abs}} + S_{JT}$, running from 3.83 to 5.62, whose Jahn-Teller-active remainder is S_{JT} (right, red). Panels (a) and (c) mark the two limits of use: the window edge, and the zero-phonon weight above 40 GPa.

- `theory_a1_generalization.py` — the coincidence condition, the exact degeneracy, the asymmetric 5% band, and the T4 inference check of Sec. VI. Its `v1_envelope_local_maxima` counts the interior local maxima of the single-effective-mode envelope against pressure, on the grid, normalization and threshold that `Kernel.local_maxima` applies to the reconstructed kernel; this is the one-against-four comparison of Sec. IIF and Sec. VC.
- `theory_a2_multiplicity.py` — Theorem M and Theorem G: Tables I, III and II, the gauge degeneracy, and the invariance of the ladder. In `ladder_is_gauge_invariant` each member of the gauge family is given its own half-scale Γ , so the five are probed at absolute intensities spanning a factor 16 and the agreement of their ladders is a result rather than an identity.
- `dreau_exponent.py` — the splitting exponent $E = 3$, $\rho^* = 1/5$, and the reachability estimate of Eq. (18).
- `theory_a3_branch_exchange.py`, `figure_validation.py` — Theorem X, the source-figure check, and the drivers of Table IV and the bandwidth-dependent $P^*(W_{\text{eff}})$ curve of Fig. 3(b).
- `kernel_sanity_checks.py` — the checks K1–K4 of

Appendix B 2.

- `repro_yield.py` — the cross-figure validation of Appendix B 1.
- `fig_g_gauge_degeneracy.py`,
`fig_m_level_set_ladder.py`,
`fig_x_branch_exchange.py`,
`esa_figv_kernel_sanity.py` — Figs. 1, 2, 3, 4 and 5. The last script draws the last two.

cd code

```
python report_120gpa_sensitivity.py
python theory_a1_generalization.py
python theory_a2_multiplicity.py
python theory_a3_branch_exchange.py
python figure_validation.py
python kernel_sanity_checks.py
python dreau_exponent.py
python fig_m_level_set_ladder.py
python fig_g_gauge_degeneracy.py
python fig_x_branch_exchange.py
python esa_figv_kernel_sanity.py
python -m pytest . -q
```

The frozen record of the theory, including the errata that withdrew the zero-phonon peak heights and the single-valued P^* , is `docs/theory/theory_freeze_v3_ho_integrated.md`; the derivations are in `docs/theory/theory_a1_numerical_execution.md`, `docs/theory/theory_a2_multiplicity.md` and `docs/theory/theory_a3_branch_exchange.md`.

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